Pedagogical Support Material

Quantum Mechanics & Quantum Chemistry Programme

For Students and Instructors

Section 1: Learning Tiers (HS → PhD)
Definitions, Competencies & Worked Example (Module I.2)

Section 2: Assessment Artifact Set — Revised Taxonomy
Undergraduate Track (SCQU, CCQU, SPSU, CPSU, SPJU, CPJU)
Graduate Track (SCQG, CCQG, SPSG, CPSG, SPJG, CPJG)
Research Track (WRQ ×5, ORQ ×5)

Section 3: Tiered Bibliography — Structure & Design Principles
(Full reading lists in companion document
QM Programme: Tiered Bibliography)

v3 — Companion to the Course Syllabus.
Sets the shared language for how *material is taught and assessed across all Series and Modules of the programme.*

See also: ***QM Programme: Tiered Bibliography*** *(separate companion document)*

Contents

Introduction

This document serves two audiences simultaneously.

For students: It explains what is expected at each stage of your education and what kinds of tasks you will be asked to perform. Understanding the tier system and the artifact taxonomy before a course begins removes ambiguity about difficulty levels and grading expectations.

For instructors: It provides a reusable, course-independent specification for how to differentiate content and assessment. The tier system and revised artifact set described here apply uniformly across *every lecture* of every module in the programme.

How the two sections connect. Section 1 defines the *audience* dimension. Section 2 defines the *assessment* dimension, now organised into three tracks:

- **Undergraduate track** (Tiers 1–3): six artifact types, each with a simple and complex variant labelled `U`
- **Graduate track** (Tiers 4–5): six parallel artifact types with deeper scope, labelled `G`
- **Research track** (Tiers 4–5): five well-defined and five open-ended research questions

1 Learning Tiers

Overview. The programme is designed for a *mixed cohort*: a lecture may be attended simultaneously by students with only secondary-school mathematics and PhD researchers extending their foundations. The five-tier system ensures every student finds appropriate entry points and stretch goals without fragmenting the cohort into separate tracks.

The tiers are *not grade labels*. They describe the prerequisite competency set a student brings. A student may operate at different tiers for different topics (e.g. Tier 3 for linear algebra, Tier 2 for group theory).

Track alignment: Tiers 1–3 correspond to the **Undergraduate track**; Tiers 4–5 correspond to the **Graduate track** and **Research track**.

1.1 Tier Ladder at a Glance

Tier	Label	Track	One-line description
1	High School (HS)	Undergrad	Amplitudes are complex arrows; probabilities are squared lengths
2	Beginning UG (BegUG)	Undergrad	Bases and components; compute probabilities with $\langle n\|\psi\rangle$
3	Advanced UG (AdvUG)	Undergrad	Inequalities; geometry and internal consistency of the formalism
4	Master's (MSc)	Graduate	Rays and projective space; symmetries via Wigner's theorem
5	PhD	Graduate	Rigorous Hilbert space; separability; domain subtleties

1.2 Tier Definitions and Competencies

Tier 1 — High School (HS) [Undergraduate track]

Goal: Build the *intuition tools* needed for amplitudes and interference.

Core competencies:

- Complex numbers as "2D arrows": $a + ib$, magnitude $|z|$, phase $\arg z$
- Magnitude squared as probability: $P \propto |A|^2$
- Vector addition analogy: adding complex amplitudes before squaring gives interference
- Basic probability: events, normalisation ("total probability = 1")

You should be able to:

- Compute $|a + ib|^2$
- Understand why $|A_1 + A_2|^2 \neq |A_1|^2 + |A_2|^2$ in general

Tier 2 — Beginning Undergraduate (BegUG) [Undergraduate track]

Goal: Become fluent in the *working language* of bras, kets, bases, and probabilities.

Core competencies:

- Normalisation: $\langle\psi|\psi\rangle = 1$
- Orthonormal bases: $\langle m|n\rangle = \delta_{mn}$
- Components as inner products: $c_n = \langle n|\psi\rangle$
- Probabilities: $P(n) = |c_n|^2$

- Insert identity (discrete): $\sum_n |n\rangle\langle n| = \hat{\mathbb{1}}$

You should be able to:
- Expand a state in a basis and compute probabilities
- Translate between $|\psi\rangle$ and its coordinate column vector in a chosen basis

Tier 3 — Advanced Undergraduate (AdvUG) [Undergraduate track]

Goal: Understand why the formalism is *internally consistent* and geometric.

Core competencies:
- Cauchy–Schwarz inequality: $|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle\langle\psi|\psi\rangle$
- Triangle inequality: $\|\phi + \psi\| \leq \|\phi\| + \|\psi\|$
- Metric/geometric viewpoint: overlaps as cosines, distance from $\|\phi - \psi\|$

You should be able to:
- Prove basic inequalities in Hilbert space
- Explain why overlaps and probabilities are bounded by 1 for normalised states
- See superposition as linear structure, not mystery

Tier 4 — Master's (MSc) [Graduate track]

Goal: Shift from "vectors as states" to *rays as states* and connect to symmetry theory.

Core competencies:
- Rays / projective Hilbert space: physical states are equivalence classes $|\psi\rangle \sim e^{i\theta}|\psi\rangle$; pure states are points in $\mathbb{P}(\mathcal{H})$
- Wigner theorem preview: transformations preserving $|\langle\phi|\psi\rangle|^2$ are unitary or antiunitary

You should be able to:
- Explain why global phase is unphysical and why rays are the correct objects
- State Wigner's theorem and why it matters for symmetries

Tier 5 — PhD [Graduate track]

Goal: Make all foundational statements *mathematically precise* and know where physics shorthand hides subtleties.

Core competencies:
- Precise definition of Hilbert space (inner product + completeness)
- Separability (countable orthonormal basis / countable dense subset)
- Rays as 1D subspaces; $\mathbb{P}(\mathcal{H})$ as a metric space
- Operator subtleties: Hermitian vs. self-adjoint; generalised eigenvectors; rigged Hilbert spaces

You should be able to:
- Distinguish informal bra-ket from rigorous operator statements
- Discuss why separability matters for spectral decompositions
- Frame state space as a projective object and symmetry as structure on that space

1.3 What Changes Across Tiers (Same Artifact, Different Depth)

Instructor note: *The same assignment can be given to all tiers simultaneously. What changes is the* language*, the required* rigor*, and the* scope*. Below: "phase invariance" across all five tiers.*

Tier	How "phase invariance" is engaged
HS	"Why doesn't the arrow's overall rotation matter for length?" — intuitive analogy
BegUG	Compute $\lvert e^{i\theta}c_n\rvert^2 = \lvert c_n\rvert^2$; verify for a 2-component example
AdvUG	Prove phase invariance for all probabilities in any orthonormal basis expansion
MSc	Well-definedness of Born-rule map on rays in $\mathbb{P}(\mathcal{H})$
PhD	Formalise as $U(1)$ fiber structure; Hopf fibration $S^1 \to S^{2n+1} \to \mathbb{CP}^n$

1.4 Worked Example: Module I.2 — Lecture L01

The following shows how **Lecture L01: *Why Dirac Notation? States as Rays & Superposition*** is differentiated across all five tiers. This pattern repeats for every lecture in the programme.

L01 topic: States, kets, rays, global phase, Born rule preview. Dual Sakurai (operational) and Dirac (abstract) perspectives.

Tier 1 (HS) — L01

Focus: $|z|^2$ as probability; interference with two paths; ket as a "direction arrow"; why global rotation of the arrow changes nothing.

Signature problem: Given $A_1 = \frac{1}{\sqrt{2}}$, $A_2 = \frac{i}{\sqrt{2}}$, compute total probability; verify it equals 1.

Tier 2 (BegUG) — L01

Focus: Normalisation; components $c_n = \langle n|\psi\rangle$; notation dictionary.

Signature problem: $|\psi\rangle = \frac{3}{5}|+\rangle + \frac{4}{5}|-\rangle$. Verify normalisation; find $P(+)$ and $P(-)$; multiply by $e^{i\pi/3}$; confirm probabilities unchanged.

Tier 3 (AdvUG) — L01

Focus: Prove global phase is exactly unobservable; distinguish global from relative phase.

Signature problem: Prove that for any normalised $|\psi\rangle$, any ONB $\{|n\rangle\}$, and any $\theta \in \mathbb{R}$: $|\langle n|e^{i\theta}|\psi\rangle|^2 = |\langle n|\psi\rangle|^2$ for all n.

Tier 4 (MSc) — L01

Focus: Rays as equivalence classes; projective Hilbert space $\mathbb{P}(\mathcal{H})$; Born-rule map well-defined on rays; preview of Wigner's theorem.

Signature problem: Define the ray equivalence relation; show Born-rule probability depends only on $[\psi]$, not the representative; frame what a "symmetry" of physical states means.

Tier 5 (PhD) — L01

Focus: Hilbert space completeness; separability; rays as 1D subspaces; Fubini–Study metric; full statement of Wigner's theorem.

Signature problem: (a) State the definition of a separable Hilbert space. (b) Why does separability guarantee a countable ONB? (c) State Wigner's theorem precisely, including the $U(1)$ lifting ambiguity. (d) Give one example of an antiunitary symmetry and explain its physical significance.

2 Assessment Artifact Set — Revised Taxonomy

Design goals. Each lecture assesses four capabilities:

1. **Conceptual understanding** — can you explain it?
2. **Procedural fluency** — can you compute it?
3. **Structural mastery** — can you prove or derive it?
4. **Transfer & creation** — can you apply it to a new context or ask new questions?

The revised taxonomy provides **parallel sets** for the undergraduate and graduate tracks, plus a shared research track, for a total of **14 artifact types** per lecture. Each type has a defined count, scope, and rubric.

2.1 Complete Artifact Taxonomy at a Glance

Code	Full Name	Track	Tiers	Count
Undergraduate Concept & Problem Artifacts				
SCQU	Simple Concept Questions — Undergrad	UG	1–3	10 questions
CCQU	Complex Concept Questions — Undergrad	UG	2–3	10 questions
SPSU	Simple Problem Set — Undergrad	UG	1–3	10 problems
CPSU	Complex Problem Set — Undergrad	UG	3	5 problems
SPJU	Simple Project — Undergrad	UG	2–3	1 project
CPJU	Complex Project — Undergrad	UG	3	1 project
Graduate Concept & Problem Artifacts				
SCQG	Simple Concept Questions — Grad	Grad	4–5	10 questions
CCQG	Complex Concept Questions — Grad	Grad	4–5	10 questions
SPSG	Simple Problem Set — Grad	Grad	4–5	10 problems
CPSG	Complex Problem Set — Grad	Grad	4–5	5 problems
SPJG	Simple Project — Grad	Grad	4–5	5 projects
CPJG	Complex Project — Grad	Grad	4–5	5 projects
Research Track (shared Tiers 4–5)				
WRQ	Well-Defined Research Question	Research	4–5	5 questions
ORQ	Open-Ended Research Question	Research	4–5	5 questions
	Total artifacts per lecture			**92 items**

2.2 Capability–Artifact Mapping

Capability	UG artifacts	Grad artifacts
Conceptual understanding	SCQU, CCQU	SCQG, CCQG
Procedural fluency	SPSU	SPSG
Structural mastery	CPSU	CPSG
Transfer (engineering)	SPJU	SPJG (×5)
Transfer (synthesis)	CPJU	CPJG (×5)
Creation / frontier	—	WRQ (×5), ORQ (×5)

2.3 Undergraduate Track Artifacts

UNDERGRADUATE TRACK (Tiers 1–3: HS, BegUG, AdvUG)

2.3.1 SCQU — Simple Concept Questions (Undergrad)

SCQU — Simple Concept Questions, Undergraduate 10 questions

Purpose: Fast checks that a Tier 1–3 student has the *right mental model*, not just algebra. Pitched at intuition and working-language levels.

Format:

- 10 questions — mix of multiple-choice (4), true/false + one-sentence justify (4), and 1–2 sentence short answer (2)
- 15–20 minutes in class or as pre-lecture warmup

Tier targeting:

HS	Qualitative only; language avoids formalism; diagrams welcome
BegUG	Qualitative + brief algebraic check (e.g. verify $\|c_1\|^2 + \|c_2\|^2 = 1$)
AdvUG	Include "identify the correct statement" from near-equivalent options

Example prompts (L01 style):

- Why is $|\psi\rangle$ not "a column vector" until a basis is chosen?
- What is physically meaningful: $\langle a|\psi\rangle$ or $|\langle a|\psi\rangle|^2$?
- True/False: multiplying $|\psi\rangle$ by $e^{i\theta}$ changes the probability of any outcome. (Justify.)
- In what sense is "superposition" basis-dependent?

SCQU Rubric

Correct answer	1.5 pts each
Brief justification consistent	0.5 pts each

Total: 20 points (10 questions $\times$ 2 pts)

2.3.2 CCQU — Complex Concept Questions (Undergrad)

CCQU — Complex Concept Questions, Undergraduate 10 questions

Purpose: Test *structural understanding* at the UG level: not just "what" but "why the formalism is consistent." Requires short arguments, not full proofs.

Format:

- 10 questions — mix of "explain why" (4), "what would change if…" (3), and "identify the flaw in this argument" (3)
- 25–35 minutes; homework or tutorial context

Tier targeting:

BegUG	Explain consequences of definitions; trace through an argument
AdvUG	Identify logical gaps; construct a counterexample; compare two approaches

Example prompts (L01 style):

- Why is the global phase of a quantum state unobservable, but the *relative* phase between components is not? Give a concrete example.
- A student claims: "$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ describe the same physical state." What is wrong with this claim?
- If Hilbert space were defined over the real numbers instead of complex numbers, which aspect of quantum mechanics would immediately fail? Justify briefly.

CCQU Rubric

Correct identification of the key point	1.0 pt each
Coherent short argument or example	1.0 pt each

Total: 20 points (10 questions × 2 pts)

2.3.3 SPSU — Simple Problem Set (Undergrad)

SPSU — Simple Problem Set, Undergraduate 10 problems

Purpose: Build *procedural fluency*: compute probabilities, normalise states, insert identity, perform basic basis changes.

Format: 10 problems; 1–5 steps each; minimal proofs.

Internal structure:

- **Part A — mechanics** (4 problems): normalisation, components, overlaps
- **Part B — interpretation** (3 problems): what does the computed number mean?
- **Part C — translation** (3 problems): ket ↔ coordinates ↔ wavefunction preview

Tier scaling:

HS	Numeric complex arithmetic + amplitude-to-probability conversion
BegUG	Normalisation + components $c_n = \langle n|\psi\rangle$ + probability sums
AdvUG	Include representation-change: verify result independent of basis choice

SPSU Rubric (per problem)

Correct result	60%
Correct method/notation	30%
Interpretation sentence	10%

2.3.4 CPSU — Complex Problem Set (Undergrad)

CPSU — Complex Problem Set, Undergraduate 5 problems

Purpose: Assess *structural mastery* at UG level: proofs, derivations, and multi-step synthesis accessible to Tier 3 students.

Format: 5 multi-part problems (a)(b)(c)...; 3–6 hours total.

Canonical themes:

- Prove Cauchy–Schwarz and/or triangle inequality
- Prove phase invariance and ray equivalence (with guidance)
- Show unitaries preserve probabilities and inner products
- Derive completeness consequences (Parseval identity)
- Construct a worked proof of the spectral theorem for a 2×2 Hermitian matrix

Tier targeting: Primarily AdvUG (Tier 3); parts (a) accessible to BegUG.

CPSU Rubric (per problem)

Correct statement of assumptions	15%
Logical proof structure	45%
Correct algebra/steps	25%
Insight / interpretation	15%

2.3.5 SPJU — Simple Project (Undergrad)

SPJU — Simple Project, Undergraduate 1 project

Purpose: Hands-on transfer: "I can build something that uses the formalism."

Scope: 2–6 hours; structured, scaffolded; works for mixed Tier 1–3 cohort.

Deliverable: Short report (1–3 pages) *and* a small artifact (code, worksheet, or visualisation).

Example — Basis Expansion Calculator (L01):

- *Inputs:* coefficients c in basis $\{|n\rangle\}$; unitary U
- *Outputs:* new coefficients $c' = Uc$
- *Required checks:* normalisation preserved; probabilities sum to 1; global phase invariance verified
- *Deliverables:* code + README; one worked example; short explanation of what changed and what did not

SPJU Rubric

Correctness	40%
Verification tests included	25%
Clarity of explanation	25%
Reproducibility (instructions)	10%

2.3.6 CPJU — Complex Project (Undergrad)

CPJU — Complex Project, Undergraduate 1 project

Purpose: Synthesis and conceptual transfer: "I can compare frameworks and defend choices."

Scope: 8–15 hours; open-ended but guided; mini-essay or mini-tool with reflection.

Deliverable: 5–8 page report + appendix with computations + optional code.

Example — Compare Representations (L01):

1. Represent a chosen state as: (i) abstract ket, (ii) coordinates in basis A, (iii) coordinates in basis B, (iv) wavefunction preview $\psi(x) = \langle x|\psi\rangle$
2. Analyse: what is invariant (norm, probabilities)? What changes (coordinates)?

CPJU Rubric

Technical correctness	30%
Conceptual synthesis	35%
Depth & structure	25%
Originality / insight	10%

2.4 Graduate Track Artifacts

GRADUATE TRACK (Tiers 4–5: MSc, PhD)

2.4.1 SCQG — Simple Concept Questions (Grad)

SCQG — Simple Concept Questions, Graduate 10 questions

Purpose: Rapid check that a Tier 4–5 student can *state and navigate* the rigorous formalism without hesitation.

Format: 10 questions; mix of short statement (4), true/false with proof sketch (3), and "identify the precise error" (3); 20–25 minutes.

Tier targeting:

MSc	Conceptual statements about rays, operators, measurement; standard results
PhD	Include precise domain/boundedness conditions; statement of named theorems

Example prompts (L01 style):

- State precisely what it means for a Hilbert space to be separable.
- True/False (with proof sketch): every Hermitian operator on $\mathcal{H}$ is self-adjoint.
- What is a ray in $\mathcal{H}$, and why are rays the correct mathematical objects for pure quantum states?
- Give the exact statement of Wigner's theorem (including the $U(1)$ ambiguity).

SCQG Rubric

Precise correct statement	1.5 pts each
Brief justification or proof sketch	0.5 pts each

Total: 20 points

2.4.2 CCQG — Complex Concept Questions (Grad)

CCQG — Complex Concept Questions, Graduate 10 questions

Purpose: Test *deep structural understanding* at graduate level: connecting formalism to mathematical foundations, identifying subtle distinctions, and engaging with open or contested points.

Format: 10 questions; mix of "distinguish and explain" (4), "what fails without assumption X" (3), and "connect these two results" (3); 35–50 minutes.

Example prompts (L01 style):

- Distinguish Hermitian from self-adjoint operators. Give an example of an operator that is one but not the other.
- Why does the Fubini–Study metric on $\mathbb{P}(\mathcal{H})$ encode transition probabilities? Show the connection explicitly.
- What additional structure, beyond a Hilbert space, is needed to define the Gel'fand triple? Why is this structure physically necessary?
- Wigner's theorem guarantees every symmetry lifts to a unitary or antiunitary operator.

What happens if we drop the assumption that the map preserves $|\langle\phi|\psi\rangle|^2$? Is there still a useful structure theorem?

CCQG Rubric

Correct identification of key distinction/connection	1.0 pt each
Rigorous short argument or counterexample	1.0 pt each

Total: 20 points

2.4.3 SPSG — Simple Problem Set (Grad)

SPSG — Simple Problem Set, Graduate 10 problems

Purpose: Build *fluency with the rigorous formalism*: operator domains, spectral theory computations, density matrices, and representation-independence checks.

Format: 10 problems; each 2–8 steps; representation-independence always verified.

Internal structure:

- **Part A — operator mechanics** (4 problems): adjoint, spectrum, domain checks, expectation values in density-matrix form
- **Part B — rigorous translation** (3 problems): continuous bases, delta distributions, Fourier connection
- **Part C — symmetry & invariance** (3 problems): commutator algebra, unitary equivalence, Wigner theorem applications

Tier scaling:

MSc	Work in density-matrix formalism; verify trace-class and positivity conditions
PhD	Additionally check domain conditions; apply functional calculus

SPSG Rubric (per problem)

Correct result with domain/condition check	50%
Correct method and rigorous notation	35%
Interpretation or physical meaning	15%

2.4.4 CPSG — Complex Problem Set (Grad)

CPSG — Complex Problem Set, Graduate 5 problems

Purpose: Assess *graduate-level structural mastery*: complete rigorous proofs, functional analysis arguments, and graduate-level derivations.

Format: 5 multi-part problems; 5–10 hours total; no guidance scaffolding.

Canonical themes:

- Prove the nuclear spectral theorem in rigged Hilbert space context
- Prove Stone's theorem (statement + sketch of key steps)
- Prove that $\hat{p} = -i\hbar d/dx$ is self-adjoint on $H^1(\mathbb{R})$ but only symmetric on $L^2([0,\infty))$

- Prove the Riesz representation theorem and connect to the bra as linear functional
- Derive the Fubini–Study metric on $\mathbb{CP}^n$ from the Hilbert space inner product

CPSG Rubric (per problem)

Precise statement of hypotheses	15%
Logical proof structure	40%
Correct analysis / functional arguments	30%
Contextual insight / connections	15%

2.4.5 SPJG — Simple Projects (Grad)

SPJG — Simple Projects, Graduate 5 projects

Purpose: Structured transfer at graduate level: "I can implement and test a rigorous idea." Five independent projects, each with a clear deliverable.

Scope per project: 3–8 hours; structured, with defined inputs and outputs.

Deliverable per project: Technical note (2–4 pages) + code or formal derivation document.

Typical SPJG project types (L01 style):

1. **Operator domain explorer:** For $\hat{p}$ on various domains, classify Hermitian/symmetric/self-adjoint; tabulate deficiency indices.
2. **Density matrix diagnostics:** Given 4×4 matrices, test $\hat{\rho} \geq 0$, $\operatorname{Tr} \hat{\rho} = 1$, purity; classify mixed/pure/entangled.
3. **Wigner function sampler:** Compute and plot the Wigner quasi-probability for a set of states; identify negativity signatures.
4. **Spectral measure calculator:** For a simple Hamiltonian, construct the spectral projection E_λ and verify the spectral theorem numerically.
5. **Ray geometry visualiser:** Implement and visualise the Fubini–Study metric for $\mathbb{CP}^1$ (Bloch sphere); compute geodesic distances between given states.

SPJG Rubric (per project)

Mathematical correctness	40%
Rigour of conditions/checks	25%
Clarity of technical note	25%
Reproducibility	10%

2.4.6 CPJG — Complex Projects (Grad)

CPJG — Complex Projects, Graduate 5 projects

Purpose: Open synthesis at graduate level: "I can connect frameworks, identify limitations, and produce research-quality output." Five independent projects, each requiring significant independent work.

Scope per project: 10–20 hours; open-ended but guided by a question.

Deliverable per project: 6–10 page technical report + appendix.

Typical CPJG project types (L01 style):

1. **Real vs. complex Hilbert space:** Identify which quantum phenomena fail in a real Hilbert space; connect to Wigner's theorem over $\mathbb{R}$.
2. **Self-adjoint extensions:** For $\hat{p}$ on $[0, 1]$ with varied boundary conditions, classify all self-adjoint extensions; compute their spectra.
3. **Rigged Hilbert space construction:** Build explicitly the Gel'fand triple $\Phi \subset L^2(\mathbb{R}) \subset \Phi'$ for the harmonic oscillator; identify the generalised eigenvectors of $\hat{x}$.
4. **Projective space and entanglement:** Characterise the image of the Segre embedding $\mathbb{CP}^1 \times \mathbb{CP}^1 \hookrightarrow \mathbb{CP}^3$; interpret the complement as entangled states.
5. **Measurement model analysis:** Compare von Neumann, Lüders, and POVM update rules; identify conditions under which they agree and diverge.

CPJG Rubric (per project)

Technical rigour and correctness	30%
Conceptual synthesis	30%
Depth, structure, and exposition	25%
Originality / critical insight	15%

2.5 Research Track Artifacts

RESEARCH TRACK (Tiers 4–5: MSc, PhD) shared across both graduate tracks

2.5.1 WRQ — Well-Defined Research Questions (×5)

WRQ — Well-Defined Research Questions 5 questions

Purpose: Train "research-like thinking" while remaining *bounded*: clear scope, clear deliverable, answerable within a defined time.

Scope per question: 5–10 hours; answerable with literature + calculation + interpretation.

Deliverable per question — 2–6 page memo:

- Question statement
- Method (how you will answer it)
- Evidence (citations + computations)
- Conclusion
- 5–10 references

Five example WRQ prompts (Module I.2, bra-ket context):

1. "How did Born's probabilistic interpretation change the meaning of the wavefunction between 1926–1927? Trace the primary sources."
2. "What exactly does 'state is a ray' buy us operationally? Provide examples where vectors fail but rays succeed."
3. "When and why did von Neumann introduce the density operator? What problem was it designed to solve that pure states cannot address?"
4. "What is the minimal set of axioms on an inner product space that forces the Born rule to take the form $P = |\langle a|\psi\rangle|^2$? Compare Gleason's theorem with the Dirac–von Neumann axioms."
5. "How does the choice of complex vs. real Hilbert space affect the structure of pure-state spaces? What operational consequence follows?"

WRQ Rubric (per question)

Clear scope and method	25%
Evidence quality	25%
Correct technical content	25%
Quality of argument	25%

2.5.2 ORQ — Open-Ended Research Questions (×5)

ORQ — Open-Ended Research Questions 5 questions

Purpose: Expose the frontier. No single correct answer; evaluates *quality of reasoning*, not correctness of conclusion.

Scope per question: 10–20 hours (or ongoing); may include proposing a new definition, comparing frameworks, or designing a thought experiment.

Deliverable per question — 4–10 page proposal-style note:

- Motivation
- What is known (brief review)
- What you propose or conjecture
- How you would test or argue it
- Risks and limitations

Five example ORQ prompts (Module I.2, bra-ket context):

1. "What minimal operational axioms force complex Hilbert space rather than real or quaternionic? Is the complex structure fundamental or emergent?"
2. "Can you formulate measurement and state update without projective collapse but still reproduce all observed statistics? What would such a framework look like mathematically?"
3. "The Gel'fand triple extends the Hilbert space to accommodate distributions. Can you propose an alternative foundation that avoids distributions entirely while retaining the continuous spectrum?"
4. "Wigner's theorem guarantees symmetries are unitary or antiunitary. Are there physically motivated generalisations (e.g. to mixed states, open systems, or approximate symmetries) that require a different structural result?"
5. "Is separability of the Hilbert space a physical assumption or merely a mathematical convenience? Design an operational test that would distinguish a separable from a non-separable state space in principle."

ORQ Rubric (per question)

Clarity and framing of the question	25%
Depth of engagement with literature	25%
Novelty and plausibility of proposal	25%
Awareness of limitations / open ends	25%

2.6 Exercise Packaging Standard

Instructor note: *Every individual problem in CPSU, CPSG, or any project type should be packaged in the four-layer format below. This supports self-study, tiered hints delivery, and automated repository assembly.*

Four-layer exercise packet (one file-set per problem):

Layer	Content
1. Problem Statement	Clean statement; no hints; the only file students receive initially
2. Suggestions / Hints	Scaffolded prompts; released after first attempt
3. Advanced Suggestions	Structural hints, alternative approaches, cross-references
4. Detailed Solution	Full derivation + commentary + common pitfall warnings

File naming convention:
`ex##_problem.tex` `ex##_hints.tex` `ex##_adv_hints.tex` `ex##_solution.tex`

2.7 Per-Lecture Minimum Bundle

Complete artifact bundle for one 90-minute lecture:

Code	Name	Track	Count
SCQU	Simple Concept Questions — UG	UG	10 questions
CCQU	Complex Concept Questions — UG	UG	10 questions
SPSU	Simple Problem Set — UG	UG	10 problems
CPSU	Complex Problem Set — UG	UG	5 problems
SPJU	Simple Project — UG	UG	1 project
CPJU	Complex Project — UG	UG	1 project
SCQG	Simple Concept Questions — Grad	Grad	10 questions
CCQG	Complex Concept Questions — Grad	Grad	10 questions
SPSG	Simple Problem Set — Grad	Grad	10 problems
CPSG	Complex Problem Set — Grad	Grad	5 problems
SPJG	Simple Projects — Grad	Grad	5 projects
CPJG	Complex Projects — Grad	Grad	5 projects
WRQ	Well-Defined Research Questions	Research	5 questions
ORQ	Open-Ended Research Questions	Research	5 questions
	Total per lecture		**92 items**

2.8 Suggested Assessment Weights

Track	Artifact group	Weight	Rationale
UG	SCQU + CCQU (concept)	15%	Mental models and structural understanding
	SPSU (fluency)	25%	Core procedural skills; highest volume
	CPSU (mastery)	30%	Proof and derivation ability
	SPJU + CPJU (transfer)	30%	Application and synthesis
Grad	SCQG + CCQG (concept)	15%	Rigorous conceptual command
	SPSG (fluency)	20%	Formal computation fluency
	CPSG (mastery)	30%	Graduate proof standards
	SPJG + CPJG (transfer)	35%	Research-quality synthesis
Research	WRQ (×5)	50%	Scoped research capacity
	ORQ (×5)	50%	Frontier reasoning

Appendix — Quick Reference Card

Tier × Artifact Eligibility Grid

	Concept Q.		**Problem Set**		**Projects**		**Concept Q.**		**Problem Set**		**Projects**		**Research**	
	SCQU	CCQU	SPSU	CPSU	SPJU	CPJU	SCQG	CCQG	SPSG	CPSG	SPJG	CPJG	WRQ	ORQ
HS	✓		✓		✓									
BegUG	✓	✓	✓		✓	opt.								
AdvUG	✓	✓	✓	✓	✓	✓								
MSc							✓	✓	✓	✓	✓	✓	✓	✓
PhD							✓	✓	✓	✓	✓	✓	✓	✓

Artifact Duration and Output Summary

Code	**Format**	**Est. time**	**Primary output**	**Pts**
SCQU	10 questions	15–20 min	Answers + brief justifications	20
CCQU	10 questions	25–35 min	Short arguments + examples	20
SPSU	10 problems	2–4 hrs	Worked solutions	100
CPSU	5 problems	3–6 hrs	Proofs + derivations	50
SPJU	1 project	2–6 hrs	Code/worksheet + 1–3 pg report	100
CPJU	1 project	8–15 hrs	5–8 pg report + appendix	100
SCQG	10 questions	20–25 min	Precise statements + sketches	20
CCQG	10 questions	35–50 min	Rigorous arguments	20
SPSG	10 problems	3–6 hrs	Formal solutions + checks	100
CPSG	5 problems	5–10 hrs	Graduate-level proofs	50
SPJG	5 projects	3–8 hrs ea.	Tech notes + code	5×100
CPJG	5 projects	10–20 hrs ea.	Full technical reports	5×100
WRQ	5 memos	5–10 hrs ea.	2–6 pg research memo	5×100
ORQ	5 proposals	10–20 hrs ea.	4–10 pg proposal note	5×100

3 Tiered Bibliography — Structure & Design Principles

This section explains how the companion bibliography document is built and why. The full reading lists — one per learning tier, plus a complete alphabetical catalogue — are published as a separate document: ***QM Programme: Tiered Bibliography***. This section gives instructors and students the conceptual framework needed to *use*, *extend*, and *adapt* that bibliography for other modules and disciplines.

3.1 Purpose of a Tiered Bibliography

A flat reading list attached to a syllabus has a fundamental problem: it collapses five very different audiences into one list. A Tier 1 student who opens Reed & Simon's *Methods of Modern Mathematical Physics* will be lost; a PhD student assigned Feynman's *QED* may feel patronised. The tiered bibliography solves this by providing **one list per tier**, each answering a single question:

What should I read next*, at* my *level, to go deeper into the material of this lecture?*

The tier structure mirrors exactly the five-tier pedagogy described in Section **??** of this document, and the reading strategy at each tier aligns with the artifact types described in Section **??**.

3.2 Four Categories of Material

Every entry in the bibliography belongs to exactly one of four categories, marked by a tag in bold square brackets:

Tag	Category and description
[N]	*Popular / narrative.* Builds intuition and historical context; no formal prerequisites. Assigned at Tier 1; useful context at Tier 2.
[T]	*Pedagogical textbook.* Structured course reading with exercises; the backbone of Tiers 1–4. These are the books students read chapter-by-chapter alongside lectures.
[P]	*Primary research paper.* Original source; accessed at Tier 3 (abstract + introduction), read fully from Tier 4; mandatory at Tier 5. The companion bibliography identifies 14 primary papers spanning quantum mechanics and nonlinear dynamics.
[M]	*Research monograph or review article.* Graduate-level and research-level consolidation; Tiers 4–5 and the Research track. Monographs are read selectively (relevant chapters); reviews are read in full as literature surveys.

3.3 Annotation System

Each entry in the tiered bibliography carries three elements beyond the citation:

Element	Meaning and use
Star rating	★★★ Essential — read in full ★★ Strongly recommended — read relevant chapters ★ Useful supplement — consult as needed
Type tag	[N] [T] [P] [M] — see categories above
Annotation	2–4 sentences linking the entry to specific programme modules and lecture codes (e.g. "Chapters 1–3 support Module I.2 Lectures L01–L05"). This is what distinguishes a tiered programme bibliography from a generic reading list: every entry is *placed* in the curriculum.

3.4 How the Tiers Map to Reading Strategies

Tier	Label	Primary material type	Reading strategy
1	HS	[N] popular books	Read cover-to-cover before the course
2	BegUG	[T] textbooks	One chapter per lecture; do all exercises
3	AdvUG	[T] + [P] first reads	Textbook main; read abstracts/intros of primary papers
4	MSc	[P] + [M] selected	Primary papers as main reading; textbooks as reference
5	PhD	[M] + [P] all	Monographs + full primary literature + arXiv

The tiers are **cumulative**: Tier k reading is assumed as background for Tier $k+1$. Instructors assigning material to a mixed cohort can use the tier badges in the bibliography to route students automatically.

3.5 Structure of the Companion Bibliography Document

The companion document ***QM Programme: Tiered Bibliography*** is organised as follows:

Part	Content
Introduction	How-to-use guide; annotation key; tier–strategy table
Section 1 (T1)	Popular science & first introductions — QM + Nonlinear Dynamics
Section 2 (T2)	Core undergraduate textbooks — QM + Mathematics support + QChem
Section 3 (T3)	Rigorous undergraduate & bridge texts — QM + primary papers (first encounter)
Section 4 (T4)	Graduate textbooks, many-body theory, quantum chemistry + key papers
Section 5 (T5)	Functional analysis monographs, advanced QM, essential PhD papers
Cat. A–F	Full alphabetical catalogue: Popular [A], UG textbooks [B], Grad textbooks [C], Monographs [D], Primary papers [E], Reviews [F]
Quick-index	Tier–category cross-index for rapid navigation

3.6 How to Extend the Bibliography to a New Module or Discipline

Instructor note: *The following protocol allows any instructor to build a compliant tiered bibliography for a new module (e.g. a new Series II or Series III module not yet covered).*

Five-step bibliography construction protocol:

1. **Identify one [N] entry per Tier 1.** Choose a popular/narrative book that gives physical intuition for the module's core concept without formalism. If none exists, a well-written Wikipedia article or a Nobel lecture can substitute.

2. **Identify two or three [T] entries for Tiers 2–3.** One standard course textbook (Tier 2) and one more rigorous text or bridge text (Tier 3). Annotate each with the specific chapters covering this module's lectures.

3. **Identify the 2–3 primary papers [P] that every Tier 3–5 student must encounter.** These are typically: the paper that first introduced the concept, the paper that proved the central theorem, and (optionally) the most-cited modern review. Annotate with which lecture code they support and which WRQ/ORQ prompts they feed into.

4. **Identify one or two [M] monographs for Tiers 4–5.** One standard graduate reference (Tier 4) and one advanced monograph or review survey (Tier 5). Annotate with chapters, not whole books.

5. **Assign tier badges and star ratings.** Use the three-star system. Essential (★★★) means: a student at this tier *must* read this to pass the Complex Problem Set (CPSU/CPSG) and Research Questions (WRQ/ORQ) for this module.

3.7 Relationship Between Bibliography and Assessment Artifacts

The tiered bibliography is not decorative — it feeds directly into the assessment artifacts defined in Section **??**. The table below makes the connections explicit:

Artifact	Track	Bibliography connection
SCQU / CCQU	UG	Students check answers against [N] and Tier 2 [T] entries
SPSU	UG	Problems drawn from exercises in starred Tier 2 [T] texts
CPSU	UG	Proofs derived from Tier 3 [T] and first [P] papers; students cite the source of each theorem they use
SPJU	UG	Implementations reproduce a result from a starred [T] text
CPJU	UG	Report situates the student's comparison within the Tier 2–3 textbook literature
SCQG / CCQG	Grad	Precise statements verified against Tier 4–5 [T] and [M] entries
CPSG	Grad	Full proofs reference Tier 5 [M] sources (Reed–Simon, Rudin, etc.)
SPJG	Grad	Each of the 5 projects implements or verifies a result from a specific [P] or [M] entry
CPJG	Grad	Each of the 5 complex projects requires a mini literature review using Cat. C–D entries
WRQ	Res.	Each of the 5 questions is directly seeded by a [P] entry; the memo must cite at least 5 items from the bibliography
ORQ	Res.	Each of the 5 questions pushes beyond the bibliography; students must identify at least 2 items *not* in the current catalogue and justify their inclusion

The ORQ's requirement to find items *not* in the bibliography is deliberate: it trains the research skill of *extending* a curated list, not just consuming it.

3.8 Cross-Discipline Transfer: The Nonlinear Dynamics Example

The companion bibliography includes cross-reference entries from the **Nonlinear Dynamics & Chaos** (logistic map) course at each tier. These serve two purposes:

- **Structural illustration:** They show concretely how the same five-tier architecture transfers to a completely different field. Gleick's *Chaos* [T1] plays the same role as Feynman's *QED* [T1]; May (1976) [P, T3] plays the same role as Born (1926) [P, T3]; de Melo & van Strien [M, T5] plays the same role as Reed & Simon [M, T5].
- **Interdisciplinary resonance:** Several quantum mechanics modules (especially Series II Modules II.7 and II.5) touch phenomena — ergodicity, invariant measures, spectral statistics — that have direct nonlinear dynamics analogues. The cross-references make these connections visible to students at the right tier.

3.9 Maintenance and Version Control

Recommended update cycle for the bibliography companion:

- **Annual review:** Check all URLs, update edition numbers, and scan arXiv/journal alerts for new papers that should replace or supplement existing [P] or [M] entries in Tiers 4–5.
- **Module release:** Whenever a new module is added to the programme, apply the five-step protocol (Section 3.6 above) to generate its bibliography entries and add them to the relevant tier sections.
- **Student feedback loop:** ORQ submissions frequently surface high-quality papers not in the catalogue. A nominated student each year can maintain a "candidates" list; the instructor reviews it and incorporates the best additions in the next annual update.
- **Version labelling:** The bibliography document carries a version number (e.g. `bib-v1.0`) separate from this PSM document. The two are linked but versioned independently so the bibliography

can be updated without triggering a full PSM revision.